

# UC Berkeley Math 128A: Homework 3

Prof. Per-Olof Persson (persson@berkeley.edu)

MATLAB usage classification: PEN (pen and paper only), CALC (use as calculator), CODE (run codes).

0. Please assign yourself a self grade on this homework (0, 1, or 2) based off the guidelines in the syllabus.
1. (2.2-5bd, CODE) The following methods are proposed to compute  $21^{1/3}$ . Rank them in order, based on their apparent speed of convergence, assuming  $p_0 = 1$ .
- b.  $p_n = p_{n-1} - \frac{p_{n-1}^3 - 21}{3p_{n-1}^2}$
- d.  $p_n = \left(\frac{21}{p_{n-1}}\right)^{1/2}$
2. (2.2-13abf, CALC to find interval and number of iterations, CODE to compute the approximations) For each of the following equations, determine an interval  $[a, b]$  on which fixed-point iteration will converge. Estimate the number of iterations necessary to obtain approximations accurate to within  $10^{-5}$ , and perform the calculations.
- a.  $x = \frac{2 - e^x + x^2}{3}$
- b.  $x = \frac{5}{x^2} + 2$
- f.  $x = 0.5(\sin x + \cos x)$
3. (2.3-6a, CALC) Use Newton's method to find solutions accurate to within  $10^{-5}$  for the following problem.
- a.  $e^x + 2^{-x} + 2 \cos x - 6 = 0$  for  $1 \leq x \leq 2$ .
4. (2.3-8a, CALC) Use the Secant method to find solutions accurate to within  $10^{-5}$  for the following problems.
- a.  $e^x + 2^{-x} + 2 \cos x - 6 = 0$  for  $1 \leq x \leq 2$
5. (2.3-18, CODE) Use Newton's method to solve the equation

$$0 = \frac{1}{2} + \frac{1}{4}x^2 - x \sin x - \frac{1}{2} \cos 2x, \quad \text{with } p_0 = \frac{\pi}{2}$$

Iterate using Newton's method until an accuracy of  $10^{-5}$  is obtained. Explain why the result seems unusual for Newton's method. Also, solve the equation with  $p_0 = 5\pi$  and  $p_0 = 10\pi$ .

6. (2.4-2a, CODE) Use Newton's method to find solutions accurate to within  $10^{-5}$  to the following problems.
- a.  $1 - 4x \cos x + 2x^2 + \cos 2x = 0$ , for  $0 \leq x \leq 1$ .
7. (2.4-4a, CODE) Repeat the previous exercise using the modified Newton's method. Is there an improvement in speed or accuracy?
8. (2.4 PEN) Consider the iteration

$$x_{k+1} = \frac{x_k}{2} + \frac{1}{x_k}, \quad k = 0, 1, \dots,$$

Assume that  $x_0 > 0$  and that the iteration converges.

1. What number does the iteration converge to?
2. What is the order of convergence?